Loss Given Default Downturn Estimation

Regulatory Requirements, Methods, and Challenges

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Regulatory Framework for Downturn LGD Estimates

- Loss Given Default (LGD) estimates under the Internal Rating-Based (IRB) framework should reflect economic downturn conditions.
- The regulatory framework for downturn LGD estimates is primarily defined through Article 181 of the CRR, the Commission Delegated Regulation (EU) 2021/930, the EBA Guidelines on Downturn LGD Estimation, and the ECB Guide to Internal Models.
- The downturn adjustment framework is intended to ensure that realised losses observed under adverse economic conditions are adequately reflected in capital requirements.
- The regulatory framework separates the downturn LGD process into two major components:
  - identification of the downturn period;
  - quantification of downturn impact on LGD estimates.

Main Components of the Downturn LGD Framework

- The downturn LGD framework follows a sequential process linking macroeconomic deterioration with realised credit losses.
- The first step is downturn period identification. This step focuses on identifying periods of adverse economic conditions relevant for the analysed exposure class or portfolio.
- The second step is downturn LGD quantification. This step measures the impact of identified adverse economic conditions on realised losses and LGD estimates.
- Although the downturn period identification process is often considered independent from the analysis of realised losses and quantification process in practice, this separation is not always possible due to the potential lagged impact of economic conditions on realised losses.
- At a minimum, interaction between these two steps can be expected when defining the appropriate lag structure for macroeconomic indicators or other portfolio-specific variables used for the purpose of downturn period identification.
- Regulatory requirements expect practitioners to demonstrate that identified downturn periods and the quantification process are economically meaningful and statistically supported, with the overall objective of ensuring that downturn LGD estimates remain appropriately conservative across economic cycles.

Downturn Period Identification

The downturn period identification process consists of several steps, including defining the nature, severity, and duration of the downturn period.

Nature

- Regulation defines the nature of an economic downturn through the selection of relevant economic indicators.
- As a initial step of downturn period identification process practitioners are required to identify indicators that are relevant for the analysed exposure type and sensitive to the economic cycle.
- The regulation specifies mandatory indicators applicable to all portfolios, including: GDP, unemployment rate, aggregate default rates, aggregate credit losses.
- Additional indicators are expected depending on the exposure class. For example:
  - residential mortgage portfolios should consider house price indices;
  - corporate portfolios may require sector-specific indicators;
  - commercial real estate portfolios should consider property prices and rental indices.
- Practitioners may introduce additional indicators to capture portfolio-specific cyclical behaviour.
- Finally, the selected set of indicators should define the economic nature of the downturn relevant for the portfolio and demonstrate a clear linkage between the identified portfolio risk drivers and the selected macroeconomic indicators.

Downturn Period Identification cont.

Severity

- The severity of an economic downturn is defined through the most severe observed values of the selected economic indicators.
- Regulation requires practitioners to analyse historical observations over an applicable time-span, with a minimum observation window of 20 years unless a longer period is needed to capture representative variability.
- For each indicator, practitioners identify the most severe 12-month value observed within the historical time-span.
- In practice, it can be observed that practitioners use data of different frequencies. For instance, some use yearly data, while others consider quarterly or monthly frequency for the purpose of downturn period identification. Accordingly, severity analysis should be adjusted to reflect the most severe 12-month period.
- Additionally, some practitioners reasonably consider that, given a minimum of 20 years of data, even for a single indicator, more than one 12-month period could potentially be identified for the purpose of downturn period identification.

Downturn Period Identification cont.

Duration

- The regulation defines downturn periods as phases that capture peaks and troughs linked to the most severe observed values of relevant indicators, reflecting the adverse part of the economic cycle.
- Following the regulatory requirements, practitioners typically identify one or more distinct downturn periods, ensuring that the selected duration is sufficiently long to cover the full deterioration phase.
- Where multiple indicators are used, each may suggest different downturn windows. However, methodological consistency is required when combining them into a single, final downturn definition.
- In cases where the most severe periods across two or more indicators are close or overlapping, they may be reasonably treated as a single consolidated downturn period for duration purposes.
- Practitioners are expected to clearly justify the chosen start and end points, and the resulting downturn period forms the basis for the estimation of downturn-adjusted LGD parameters.

Downturn LGD Estimation

For the purpose of LGD downturn estimation, regulation distinguishes between cases where the estimation can be performed based on observed impact, estimated impact, or when it is not possible to link realised losses with the identified downturn process.

LGD Downturn Estimation Based on Observed Impact

- Regulation identifies the observed impact approach as the preferred method for downturn LGD estimation when sufficient downturn observations are available, as it relies on direct empirical evidence from realised data.
- Under this approach, practitioners compare realised LGDs observed during downturn periods with long-run average LGDs to quantify deterioration in recoveries under adverse economic conditions.
- Practitioners are expected to demonstrate that downturn periods include a sufficient number of defaults and recoveries, and that the observed deterioration is statistically meaningful and representative of stressed conditions.
- The approach is generally considered the most reliable and least assumption-driven, as it minimises modelling uncertainty by relying on directly observed loss experience.
- However, its application may be limited in low-default or data-scarce portfolios, where downturn evidence is insufficient, in which case it is often complemented with additional conservatism adjustments.

Downturn LGD Estimation cont.

LGD Downturn Estimation Based on Estimated Impact

- When it is not possible to estimated downturn LGD based on observed impact, regulation allows for quantification on two estimation-based approaches:
  - extrapolation approach;
  - haircut approach.

LGD Downturn Estimation Based on Estimated Impact: Extrapolation Approach

- The regulation allows the use of extrapolation methods to estimate downturn impact when direct downturn observations are insufficient, using observed relationships between macroeconomic conditions and realised LGDs outside identified downturn periods.
- This approach relies on econometric modelling, macroeconomic sensitivity analysis, and scenario-based estimation to project how LGDs would behave under identified downturn conditions.
- When opting for this approach, practitioners are expected to clearly justify and document key modelling choices, including the selection of explanatory variables, model stability, and sensitivity to adverse economic conditions.
- Given the inherent model risk and parameter uncertainty, supervisory expectations generally require additional conservatism, making the approach particularly relevant for portfolios with limited downturn recovery data but sufficient cyclical information.

Downturn LGD Estimation cont.

LGD Downturn Estimation Based on Estimated Impact: Haircut Approach

- Typically in practice, haircut approaches are positioned as a fallback methodology within the observed and estimated downturn LGD framework.
- Under this approach, practitioners apply conservative adjustments to recovery rates, collateral values, or directly to LGD estimates to reflect stressed conditions.
- When using this method, practitioners are expected to ensure that applied haircuts are conservative, well-justified, and explicitly linked to downturn conditions such as declines in collateral values, weaker recovery processes, and reduced market liquidity.
- As haircut methods are considered the least risk-sensitive approach, practitioners are generally expected to apply stronger conservatism and provide enhanced expert judgment and documentation supporting the estimation process.

Downturn LGD Estimation cont.

LGD Downturn Estimation Based on Estimated Impact: Alternative Method

- The regulation requires that where downturn LGD cannot be calibrated using observed or estimation-based impact, practitioners may apply an alternative method, provided it is properly constrained and justified.
- In such cases, practitioners must ensure that a strictly positive Margin of Conservatism (MoC) of Category A is applied to reflect missing data, as part of the overall model uncertainty adjustment framework.
- Additionally, the resulting downturn LGD, including all MoC adjustments, must be greater than or equal to the minimum of long-run average LGD plus 15 percentage points and 105%, thereby ensuring a binding conservative floor.
- When using this approach, practitioners are typically expected to justify why standard observed and estimation-based approaches cannot be applied for the relevant downturn period.

Multiple Downturn Periods

- In accordance with regulatory requirements for downturn period identification, multiple downturn periods may be identified.
- In such cases, the regulation requires the following sequence of steps:
  - Calibrate a downturn LGD for each identified downturn period.
  - For each downturn period, apply the resulting downturn LGD estimates to the current non-defaulted exposures of the relevant exposure type at the time of calibration.
  - Select as the final relevant downturn period the one that results in the highest average downturn LGD, including the final MoC.
- If, for one of the identified downturn periods, the downturn LGD can only be calibrated using alternative methods, a MoC of Category A is expected to be added to the final LGD estimates calibrated using the other available downturn calibration methods.

Reference Value

- The regulation requires practitioners to compute a downturn LGD reference value as a benchmark for comparison with final downturn LGD estimates, using a structured two-step approach based on realised losses.
- First, practitioners must identify the two individual years with the highest realised losses, using all available data and ranking years by default year. Both closed and incomplete recovery processes must be included, with losses from incomplete recoveries treated consistently with the long-run average LGD methodology.
- Second, for each of the selected years, a default-weighted average realised LGD must be calculated, again incorporating both closed and appropriately treated incomplete recoveries. The final reference value is then computed as the simple average of these two annual LGD averages.
- As the reference value serves as an empirical benchmark against which final downturn LGD estimates are assessed, where material differences arise (before MoC), practitioners are expected to analyse whether these are driven by potential misidentification of downturn periods or by idiosyncratic loss behaviour in the selected years. In particular, no unidentified downturn is generally implied where the selected years overlap with the identified downturn period(s) or where the economic conditions in those years are not truly adverse. Instead, material deviations must be justified by specific loss idiosyncrasies, and practitioners should also consider potential time lags between downturn conditions and their manifestation in loss data when performing this assessment.